Extreme environmental conditions and their role in carbon dioxide and methane fluxes in forested wetlands.

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Context

- There is an increase in the frequency and intensity of extreme conditions (Seneviratne et al., 2021, IPCC).
- Analyses of 22 sites of FLUXNET-CH_4, showed that extreme CH_4 flux days make up 10–40% of the overall seasonal fluxes (Feron et al., 2024, GCB).
- Temperature and water related factors can amplify CH_4 emissions (Wang et al., 2026, AFM).
- Hydrological processes need to be better characterized in models to advance predictions of wetland CH_4 emissions (Feng et al., 2020, GRL).
- Few studies link and/or model the role of extreme weather conditions on CH_4 flux in the forested wetlands (Lippmann et al., 2025, GBC)

Research questions

- What drives high, sporadic CH_4 fluxes?
- How strong is the role of extreme events, such as heavy precipitation, in generating higher CH_4 fluxes? Are there lags? What is the magnitude of their impact?
- Are these seemingly random higher fluxes the new normal, or just outliers that we can ignore?

Approach / Methods

Three approaches were taken to understand the relationship between extreme environmental condition and extreme CH_4 fluxes.

A) Literature-based evidence
B) Analysis of flux tower data at individual sites
C) Model the processes and magnitude of change

Table 1: Site description (used in this poster presentation)

Site ID	Site Name	Location	Lat, Lon	Start – end year	Site classification
US-PFa	Park Falls/WLEF	Wisconsin	45.94, 90.27	2010-2024 (Ameriflux)	Upland
US- Esm	Everglades	Florida	25.43,80.59	2008- 2024 (Ameriflux)	Marsh /Swamp



Figure 1: Representative images of study sites US-PFa and US-Esm respectively

A. Existing Metadata/Literature

- FLUXNET-CH4 data show increasing days of extreme CH_4 flux over four decades, with compound events (e.g., hot and dry) having greater impacts. (Lippmann et al., 2025, GBC).
- CH_4 flux magnitude increases more in summer for wetlands with aerenchymatous plants, and in non-growing seasons (fall/spring) with Sphagnum moss (Feron et al., 2024, GCB).

B.1. What is the trend and magnitude of extreme CH_4 flux?

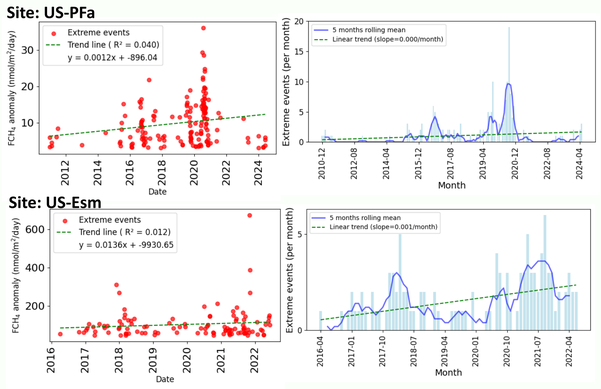


Figure 2. The anomaly of CH_4 flux (left) and the number of extremes for each month (right). Five-month moving average (blue line) and linear trend (green dotted line).

- The 95th percentile of the daily CH_4 flux values computed for each month from each site for the 2015–2025 period, showed that both extreme events and extreme CH_4 fluxes are increasing with time.

B.2. How significant is the lag between environmental variables and CH_4 flux?

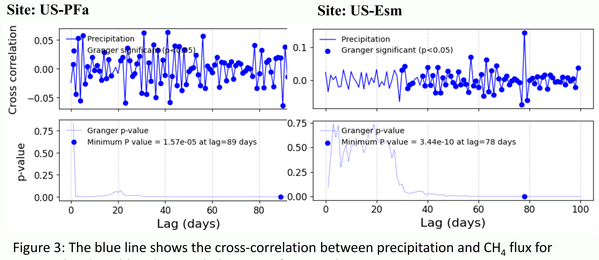


Figure 3: The blue line shows the cross-correlation between precipitation and CH_4 flux for positive lag days; blue dots mark those significant in the Granger analysis.

- While cross correlation helped to identify the strength and direction of time lagged associations, Granger causality test was useful to find the causality.
- Precipitation shows a positive predictive lag of nearly 89 days (US-Esm) indicating slower hydrological and soil responses, in these forested wetlands.

B.3. Are storms followed by CH_4 flux spikes? What is the average lag ?

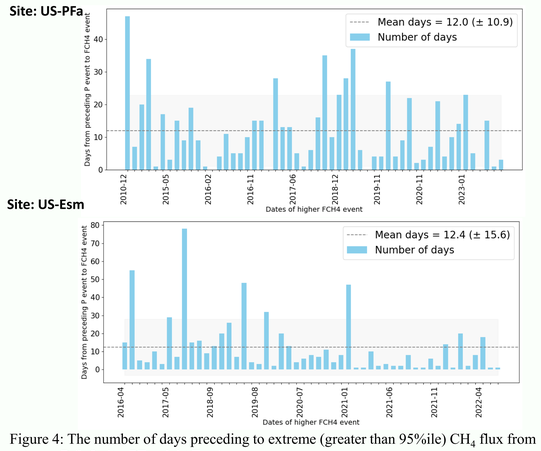


Figure 4: The number of days preceding to extreme (greater than 95%ile) CH_4 flux from extreme P (left site: PFa, right Esm)

- There was a surge in CH_4 flux 12 days after heavy storm events. However, individual events showed substantial temporal variability. In some cases an extreme CH_4 flux occurred within a few days, while others occurred nearly 75 days after a storm.

B.4. How do severe environmental conditions influence CH_4 fluxes?

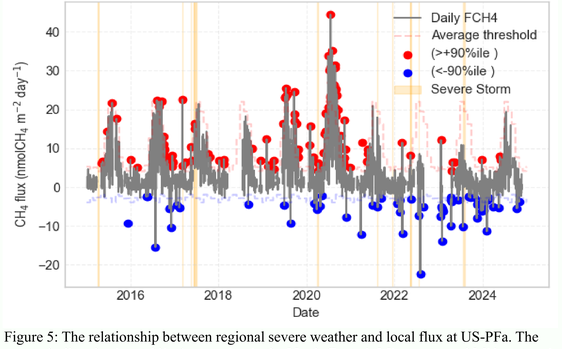


Figure 5: The relationship between regional severe weather and local flux at US-PFa. The events (yellow horizontal line) occurring in the North Central, Minnesota, and Midwest regions were considered regional events.

- Based on NOAA's Billion Dollar Weather and Climate Disasters data on severe weather events occurring in the region, there was not enough evidence to show that storms were followed by spikes in CH_4 fluxes.

C. What is the potential magnitude of change in CH_4 flux in future scenarios?

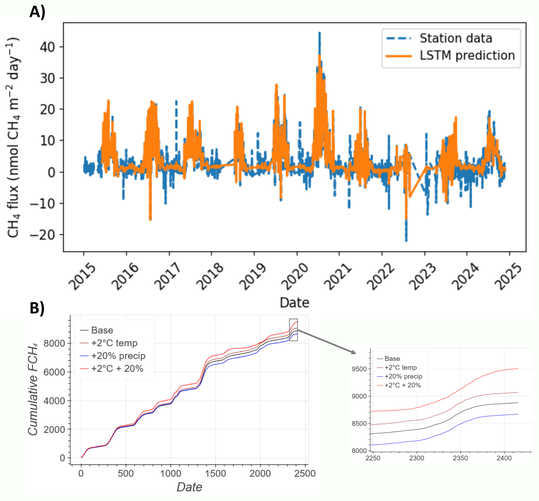


Figure 6: (A) The observed (blue) and modeled (orange) CH_4 flux pattern, in the site US-PFa. (B) The magnitude of the cumulative impact on CH_4 flux under a hypothetical scenario of increased temperature (by 2°C) and precipitation (by 20 %)

- The LSTM (Long Short-Term Memory) model effectively captures average patterns but misses some of the extremes.
- Among the scenarios, a hypothetical scenario combining temperature rise and precipitation increase showed a greater impact in terms of magnitude.

Take home message

CH_4 flux magnitude and intensity are increasing, and most probably driven by concurrent increases in other extreme environmental factors, such as recent intense storms.

Way forward

- Model the combined effect of heavy precipitation and high temperature on extreme CH_4 flux.
- Improve the LSTM model.
- Extend the study to understand net ecosystem exchange in the severe weather condition.

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